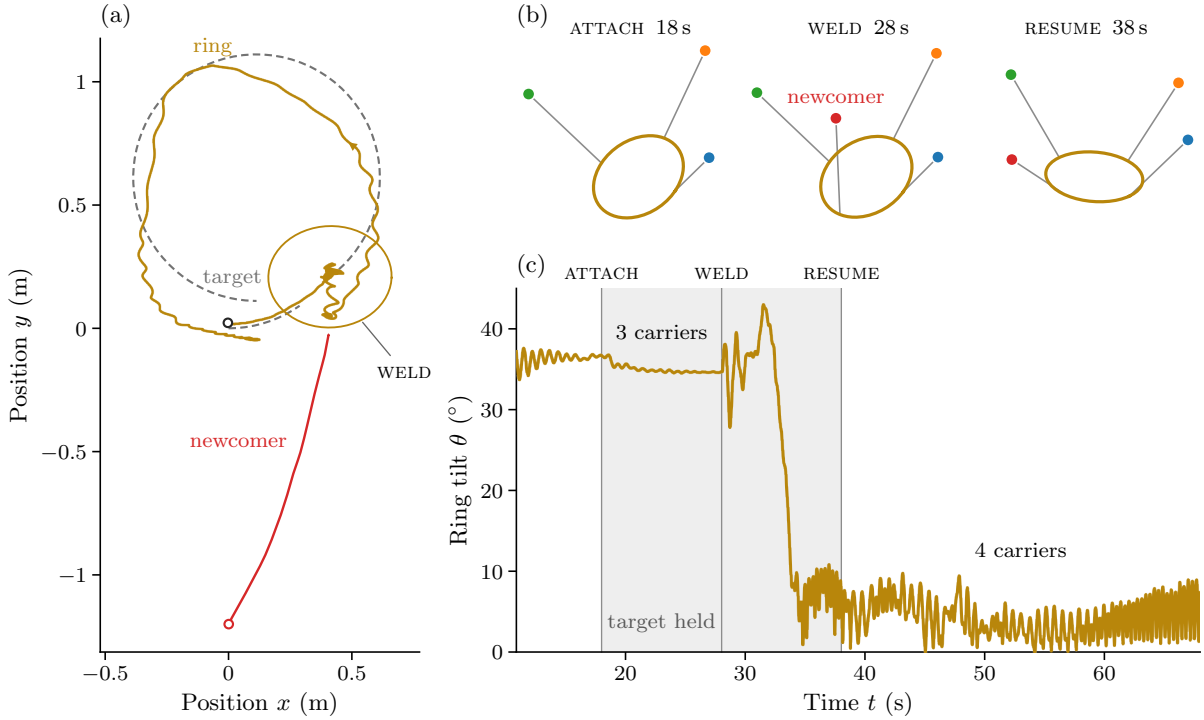

(a)
ring
target
WELD
newcomer
Position y (m)
Position x (m)
1
0.5
0
−0.5
−1
−0.5
0
0.5
(b)
ATTACH 18 s
WELD 28 s
RESUME 38 s
newcomer
(c)
ATTACH
WELD
RESUME
3 carriers
4 carriers
target held
Ring tilt θ (°)
Time t (s)
40
30
20
10
0
20
30
40
50
60